MARWAN TAREK

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SUMMARY

Software Engineer with 3+ years of experience building reliable, performance-aware software in **modern C++** (**C++17**, **STL**, templates, **RAII**, **multithreading**) and **C** on **Linux** and **Windows**, using **CMake**, **GCC/Clang/MSVC**, **GDB** and **Valgrind**. Strong in **OOP**, **design patterns**, **SOLID**, **data structures and algorithms**, unit and integration testing, and **CI/CD**. Domain experience in safety-critical railway signalling (Alstom) and automotive embedded software (Valeo), plus **Python**, **C#** and LLM tooling.

EXPERIENCE

Alstom — Software Engineer Jul 2025 – Present *Cairo, Egypt*

- Develop mixed **C++/.NET** components of safety-critical train supervision software (CBTC, ATS), built with **CMake**, **vcpkg** and **MSVC/Clang** on **Windows** and **Linux** through **GitLab CI/CD**.
- Built a **C#/.NET** service that decodes, validates and routes **Protobuf** alarms and train-status messages over **AMQP** (**ActiveMQ Artemis**), with a playback mode for deterministic tests.
- Automated **requirements-based testing**: generated test cases from system requirement specifications and measured **requirement coverage** and **traceability** across system and product levels.
- Added **Robot Framework** suites and quality gates to **GitLab CI/Jenkins**: **+30%** regression coverage.
- Built **Python** tools, a **GitLab/DOORS** SDLC dashboard and an **LLM/RAG** root-cause-analysis copilot.

Spark Systems — Software Engineer (.NET) Jan 2025 – Jun 2025 *Giza, Egypt*

- Built **C#/ASP.NET Core** services using Entity Framework, Clean Architecture and **SOLID**.

Valeo — Embedded Software Engineer Jul 2022 – Oct 2023 *Smart Village, Egypt*

- Developed and validated **multithreaded C/C++** modules for **ADAS** ECUs (LiDAR/camera): bootloader, memory and communication drivers (**CAN**, **SPI**, **I2C**, **UART**) on an **RTOS**/HAL, including **UDS** flashing.
- Ran **SiL/HiL** tests (**Vector CANoe**, **vTESTstudio**, **DOORS**) and debugged over **JTAG/SWD**; configured **AUTOSAR** (DaVinci) under **MISRA C**, **ISO 26262** and **ASPICE**.

PROJECTS (GITHUB)

- **UDS Diagnostics Client** (C++17, CMake): diagnostics library and CLI with a pluggable transport layer (mock, Linux SocketCAN ISO-TP), JSON/CSV reports and **CTest** unit tests.
- **Packet I/O Playground** (C/C++, Linux): multi-threaded packet capture with **TPACKET v3** ring buffers and fanout, PCAP/PCAPNG writers, replay and flow analysis, profiled with **perf** and **Valgrind**.
- **Dual-Bank CAN Bootloader** (C, STM32F4): A/B boot over **UDS/ISO-TP** with CRC32 manifests, atomic swap and rollback, unit-tested on a host flash simulator; Python flasher.
- **STM32F4 Drivers** (C): register-level drivers without HAL; **MISRA Evidence Pack** (Cppcheck): static analysis in CI; **UDS Test Bench** (Robot Framework): CI test suites.

SKILLS

C++: C++17, STL, templates, RAII, memory management, multithreading, atomics
Languages: C, Python, C#, Bash, SQL
Fundamentals: OOP, design patterns, SOLID, data structures & algorithms, socket programming, debugging
Tools: GCC/Clang/MSVC, Visual Studio, CMake, vcpkg, GDB, Valgrind, perf, Git, GitLab CI, Jenkins, Linux
Testing: unit/integration testing, CTest, PyTest, Robot Framework, static analysis, code coverage, SiL/HiL
Embedded & Automotive: STM32 (Cortex-M), FreeRTOS, CAN, UDS, AUTOSAR, MISRA C, ISO 26262
Also: ActiveMQ Artemis (AMQP), Kafka, Protobuf, REST, Docker, LLMs, RAG, AI agents, Agile/Scrum

EDUCATION

B.Sc. in Computer Science (Honours), Helwan University Sep 2019 – Jul 2023
Languages: English (C2), French (A2), Arabic (Native)